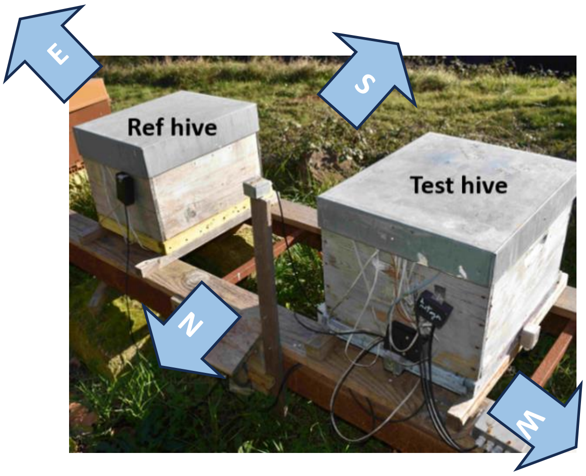
Metadata – Dadant Hive – Series « 2024_06_20_Csv »

Datasets: 2024_06_20_Environment.csv, 2024_06_20_RefHive.csv, 2024_06_20_TestHive.csv
Apiary location: Mas Lafont, 30170 CROS, France
Hive type: Empty wooden Dadant 10 frames (without frames, without bees).
Hive environment: open area
Hives orientation: Bees entrance toward south.

Description:



Two wooden Dadant hives are monitored for several days to analyse their thermal behaviour. Usually, the reference hive (Ref hive) remains unmodified during the experiments whereas the test hive undergoes several successive modifications to measure the induced effect on temperature levels. However, in this case, both hives are not modified.

Test Hive:

- 5 sensors installed on hive roof:
  north, south, east, west and central position
- 5 sensors installed on the overframe:
  north, south, east, west and central position
- 1 sensor in the upper part of the hive body
- 1 sensor in the lower part of the hive body
- 1 sensor below hive

Reference Hive:

- 1 sensor installed on hive roof (central position)
- 1 sensor installed on overframe (central position)
- 1 sensor in the upper part of the hive body
- 1 sensor in the lower part of the hive body
- 1 sensor below hive

Environment:

- Luminosity is measured on each of the 6 external sides of the hive (6 sensors).
- Outside air sensor
- Wind speed

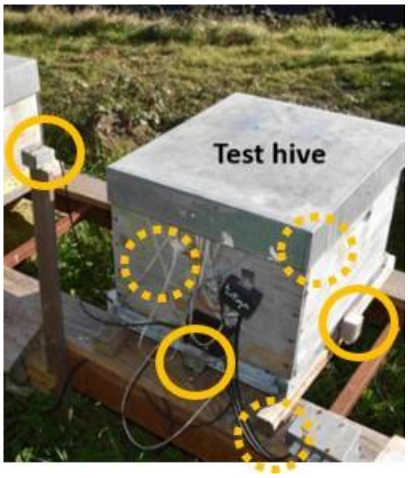


Figure 1: Luminosity sensors

Communications between the acquisition system and the sensors are done via the I2C protocol. If necessary, I2C hubs were inserted to extend the number of sensors that can be connected simultaneously.

Figure 2: I2C hubs

The sensors provide an internal analog-to-digital conversion. The sensors were individually calibrated to cancel the bias error between them to allow more precise temperature difference measurements (<0.5°C). Absolute temperature errors are probably around 1°C. The intercalibration were achieved by putting all sensors together in an insulated box.

The acquisition period is set to one minute except for the wind speed data which are acquired every 2 s and averaged over 1 min.

Data are stored using a csv format in 2 different files:

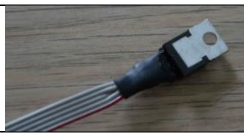
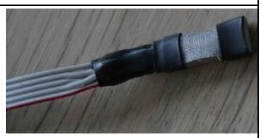
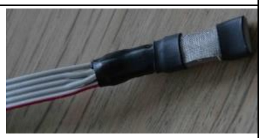
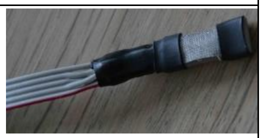
1) 2024_06_20_Environment.csv

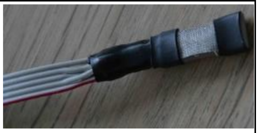
2) 2024_06_20_RefHive.csv

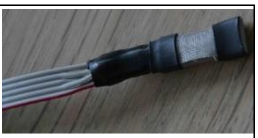
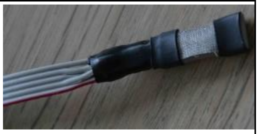
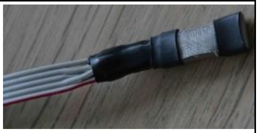
3) 2024_06_20_TestHive.csv

Comment lines are prefixed with hash character ‘#’.

Field name	Description	Unit	Sensor type	Sensor location	
Timestamp	Timestamp of measurement Year/month/ Day/Hour/ min/sec	Local time	PCF8523 (Real-time clock)+ DS3231 (Real-time clock module)	Acquisition system	
Epoch [day]	Timestamp of measurement	Day since 1 January 0			
Epoch [day]	Timestamp of measurement	Day since 1 January 0			
Wind.speed [m/s]	Wind speed	m/s	Sparkfun meteo system	Close to the apiary, 2m high	
Ground.T [deg]	Temperature at the ground surface (≈1cm)	°C	NCT75	On the ground in a shaded area.	
Sky.T [deg]	Equilibrium temperature of a horizontal black steel plane exposed to sunlight and insulated at the rear face	°C	NCT75	Fixed to the weather station	 
Test.Luminosity.south.als [lux]	Luminosity south	Lux	VEML7700 (als channel)	South side of the Dadant hive	
Test.Luminosity.south.white [lux]	Luminosity south	Lux	VEML7700 (white channel)	South side of the Dadant hive	
Test.Luminosity.top.als [lux]	Luminosity top	Lux	VEML7700 (als channel)	Top side of the Dadant hive	
Test.Luminosity.top.white [lux]	Luminosity top	Lux	VEML7700 (white channel)	Top side of the Dadant hive	
Test.Luminosity.north.als [lux]	Luminosity north	Lux	VEML7700 (als channel)	North side of the Dadant hive	
Test.luminosity.north.white [lux]	Luminosity north	Lux	VEML7700 (white channel)	North side of the Dadant hive	
Test.Luminosity.west.als [lux]	Luminosity west	Lux	VEML7700 (als channel)	West side of the Dadant hive	
Test.Luminosity.west.white [lux]	Luminosity west	Lux	VEML7700 (white channel)	West side of the Dadant hive	
Test.Luminosity.below.als [lux]	Luminosity below	Lux	VEML7700 (als channel)	Downward facing below the Dadant hive	

Test.Luminosity.below.white [lux]	Luminosity below	Lux	VEML7700 (white channel)	Downward facing below the Dadant hive	
Test.Luminosity.east.als [lux]	Luminosity east	Lux	VEML7700 (als channel)	East side of the Dadant hive	
Test.Luminosity.east.white [lux]	Luminosity east	Lux	VEML7700 (white channel)	East side of the Dadant hive	
Luminosity.horiz.ALS [lux]	Luminosity horizontal plane	Lux	VEML7700 (als channel)	Upward facing fixed on the weather station	
Luminosity.horiz.WHITE [lux]	Luminosity horizontal plane	Lux	VEML7700 (white channel)	Upward facing fixed on the weather station	
Luminosity.45.CH0 [lux]	Luminosity on a 45-tilted plane (azimuth=south)	Lux	TSL2561	Fixed on the weather station	
Luminosity.45.CH1 [lux]	Luminosity on a 45-tilted plane (azimuth=south)	Lux	TSL2561	Fixed on the weather station	
Luminosity.45.LUX [lux]	Luminosity on a 45-tilted plane (azimuth=south)	Lux	TSL2561	Fixed on the weather station	
Ref.roof [deg]	Temperature measured on Refhive roof	°C	TC74	Glued under the roof, centered	
Ref.overframe [deg]	Temperature measured on RefHive overframe	°C	NCT75	Center of the overframe	
Ref.body.top.HR [%]	Humidity measured in upper part of RefHive body	%	HIH8100	Inside hive body (top), in the center	
Ref.body.top.T [deg]	Temperature measured in upper part of RefHive body	°C	HIH8100	Inside hive body (top), in the center	
Ref.body.bottom.T [deg]	Temperature measured in lower part of RefHive body	°C	HIH8100	Inside hive body (bottom), in the center	
Ref.body.bottom.HR [%]	Humidity measured in lower part of RefHive body	%	HIH8100	Inside hive body (bottom), in the center	
Ref.below [deg]	Temperature measured below the RefHive	°C	NCT75	Centered below the hive floor	

Test.roof [deg]	Temperature measured on TestHive roof	°C	TC74	Glued under the roof, centered	
Test.roof.north [deg]	Temperature measured on the north side of the TestHive roof	°C	TC74	Hive roof north	
Test.roof.south [deg]	Temperature measured on the south side of the TestHive roof	°C	TC74	Hive roof south	
Test.roof.east [deg]	Temperature measured on the east side of the TestHive roof	°C	TC74	Hive roof east	
Test.roof.west [deg]	Temperature measured on the west side of the TestHive roof	°C	TC74	Hive roof west	
Test.overframe [deg]	temperature measured on TestHive overframe	°C	NCT75	Center of the overframe	
Test.overframe.north [deg]	Temperature measured on the north side of the TestHive overframe	°C	NCT75	Overframe north	
Test.overframe.south [deg]	Temperature measured on the south side of the TestHive overframe	°C	NCT75	Overframe south	
Test.overframe.west [deg]	Temperature measured on the west side of the TestHive overframe	°C	NCT75	Overframe west	
Test.overframe.east [deg]	Temperature measured on the east side of the TestHive overframe	°C	NCT75	Overframe east	
Test.body.top.HR [%]	Humidity measured in upper part of TestHive body	%	HIH8100	Inside hive body (top), in the center	

Test.body.top.T [deg]	Temperature measured in upper part of TestHive body	°C	HIH8100	Inside hive body (top), in the center	
Test.body.bottom.HR [%]	Humidity measured in lower part of TestHive body	%	HIH8100	Inside hive body (bottom), in the center	
Test.body.bottom.T [deg]	Temperature measured in lower part of TestHive body	°C	HIH8100	Inside hive body (bottom), in the center	
Test.below [deg]	Temperature measured below the TestHive	°C	NCT75	Centered below the hive floor